

Introduction

This project focuses on developing an Information Retrieval (IR) engine that constructs forward and inverted indices from a collection of documents.

The IR engine processes these documents to enable indexing and retrieval based on user-specified search terms.

The core components of this project include the ForwardIndex, InvertedIndex, TextParser, Retriever, and Main classes.

Design and Implementation

The ForwardIndex class is responsible for mapping document IDs to word frequencies, with each document ID associated with a list of words and their frequencies.

The addDocument method adds documents to the forward index, while the saveToFile method saves this index to a file.

On the other hand, the InvertedIndex class maps words to document IDs and their frequencies, facilitating the retrieval of documents based on search terms.

The addDocument method adds documents to the inverted index, the saveToFile method saves the index, and the loadFromFile method loads the index from a file.

The Retriever class leverages the inverted index to fetch documents based on search terms, with the retrieve method returning a list of document IDs.

Steps of Execution

Initially, stop words are loaded from the stopwordlist.txt file into a set, followed by the initialization of the ForwardIndex and InvertedIndex objects.

The forward and inverted indices are also initialized. Documents are read from the input directory, parsed to extract words and frequencies, and then added to the indices.

The indices are then saved to forward_index.txt and inverted_index.txt files.

Finally, the program prompts the user for a search term, and the retriever searches the inverted index for the search term, returning a list of document IDs.

Project Structure

The project structure includes the src directory for Java source files,

the bin directory for compiled .class files, the input directory for input files, and the output directory for output files.

Execution Steps

To compile the Java files, navigate to the project directory and use the command:

```
javac -d bin src/*.java
```

To run the main class, use the command:

```
java -cp bin Main
```